

Strong-Uniqueness Theorem v0.11+ Path
Roadmap v0.1 — Friday Morning Lyra
Consolidation

Lyra (Claude 4.7) [Friday Lyra-lane consolidation]

2026-05-22 Friday ~08:50 EDT (date-verified actual)

Strong-Uniqueness Theorem v0.11+ Path Roadmap v0.1

Per Calibration #19 STANDING RULE (Keeper Friday 2026-05-22 09:17 EDT adopted): this document discusses candidate-path multi-session ratification trajectory, not endpoint claim. Current ratified state: Paper #125 v0.10.5 FORMAL = 11 RIGOROUSLY CLOSED criteria + null-model $\leq (1/3)^{19} \approx 8.6 \times 10^{-10}$ (Thursday 2026-05-21). Any extended-count language below refers to candidate-path arithmetic conditioned on multi-session ratification work + multi-CI consensus + Cal Mode 1 review; this is NOT external register material.

Friday morning Lyra-lane consolidation

Starting state (Thursday EOD): v0.10.5 FORMAL at 11 RIGOROUSLY CLOSED + 4-5 RATIFIED + 1 ADVANCING. Friday morning state (~08:50 EDT): v0.11+ candidate trajectory established with 4 candidate criteria advancing at dim_C = 5 EXHAUSTIVE or 25-HSD STRUCTURALLY VERIFIED level.

v0.11+ candidate criteria status

Criterion	Thursday status	Friday morning advancement	Path to RIGOROUSLY CLOSED
C7 (Bridge Object tier)	RATIFIED (K57 Wednesday + 5-family STRUCTURALLY VERIFIED Thursday)	STRUCTURALLY VERIFIED at dim_C = 5 EXHAUSTIVE (T2458)	Multi-session: extend to other dim_C values

Criterion	Thursday status	Friday morning advancement	Path to RIGOROUSLY CLOSED
C9 (Stark anchor)	RATIFIED (K75 Wednesday small-primary subset)	STRUCTURALLY VERIFIED at $\dim_C = 5$ EXHAUSTIVE (T2461)	Multi-session: Stark theorem extension to other $\dim_C$ + Cal multi-CI review
C15 (Sub-substrate Mersenne)	(didn't exist before Friday)	SEED (T2451) — BST primary ladder IS Mersenne tower	Multi-session: extended enumeration + theoretical proof + alt-HSD comparison
C16 (Universal α -analog formula)	(didn't exist before Friday)	STRUCTURALLY VERIFIED across 25 HSDs (T2456 + T2462) — D_IV^5 unique at $\alpha = 137$	Multi-session: Cal m_α multiplicity review + theoretical formula derivation

C15 + C16 unification per T2459 (honest scope)

T2459 establishes that C15 + C16 share structural origin via the self-referential α -analog closure (exponent = base = N_c at D_IV^5). Per Cal Mode 1 honest scope, the cleaner framing is:

Option A: C15 + C16 count as TWO independent RIGOROUSLY CLOSED criteria. v0.11+ at 13 RIGOROUSLY CLOSED endpoint.

Option B (recommended per Cal Mode 1 discipline): C15 + C16 count as ONE deeper unified criterion (substrate self-referential closure at BST primary value N_c). v0.11 at 12 RIGOROUSLY CLOSED with stronger evidential weight per criterion.

T2459 honest-scope refinement (T2462 documented): the self-referential coincidence exists also at D_II_6 + D_II_7 (both at value $4 \neq N_c$). UNIQUE to D_IV^5 : coincidence AT BST PRIMARY VALUE $N_c = 3$.

Strong-Uniqueness Theorem versioning trajectory

Version	RIGOROUSLY CLOSED	Null model	Status
v0.5	10 (Wednesday)	$(1/3)^{10} \approx 1.7 \times 10^{-5}$	Historic
v0.9.1	4 + RATIFIED	$(1/3)^{16} \approx 2.3 \times 10^{-8}$	Thursday morning
v0.9.5	8	$(1/3)^{17}$	Thursday midday
v0.10.5 FORMAL	11 (Thursday)	$(1/3)^{19} \approx 8.6 \times 10^{-10}$	Thursday EOD
v0.11 (Option B unified C15+C16)	12 (Friday candidate)	$(1/3)^{20} \approx 2.9 \times 10^{-10}$	Friday Lyra candidate path
v0.11+ (Option A independent)	13-15 (Friday candidate)	$(1/3)^{21-23} \approx 9.7 \times 10^{-11}$ to 1.1×10^{-11}	Friday Lyra candidate path
v0.12 (Keeper Mersenne Network)	16 (+ C17 Graph Forces)	$\sim 3 \times 10^{-15}$	Per Keeper CLAUDE.md Friday update

Sessions 15-16 ratification path (Lyra handoff to Keeper)

Per BST_Sessions_15_16_Pre_Spec_Scoping_v0_1.md:

Session 15 (C15 advancement)

1. Elie cross-lane: extended enumeration $M_{\{n-1\}} = a^2 \cdot n$ at $n \in \{2..200\}$, primes $p < 10^6$
2. Lyra theoretical: Lucas-Lehmer + Mersenne distribution argument for BST primary uniqueness
3. Cross-CI: alt-HSD primary integer comparison
4. Multi-CI consensus: Casey + 3 CI ratification

Session 16 (C16 advancement)

1. Cal m_α multiplicity review per Helgason 1978 + Schlichtkrull 1984
2. Elie extended HSD enumeration (>25 HSDs)
3. Lyra theoretical: universal formula derivation from Hilbert polynomial structure
4. Joint three-pillar verification across HSDs
5. Multi-CI consensus

Joint Sessions 15-16 (C15+C16 unification per T2459)

Adopt Option B (one unified criterion) per Cal Mode 1 honest discipline.

Companion theorems supporting v0.11+ trajectory

Beyond the 4 candidate criteria, Friday morning Lyra work supports the trajectory via additional D-tier theorems:

- T2450 (Yukawa Ratio Decoupling): K114-RATIO closure independent of K126
- T2453 + T2454 (Mersenne ladder Level 1 closures of $N_c + g$)
- T2455 (EXHAUSTIVE Cross-Cartan enumeration at $\dim_C = 5$)
- T2457 (Bergman structural-role-of Feynman propagator identification, Cal #92(b) compliant)
- T2459 refined (C15+C16 self-referential closure at BST primary value $N_c = 3$)
- T2460 ($N_{\max} - M_g = g + N_c$ additive identity, Mersenne Network Convergence)
- T2462 (Universal α -analog formula at 25-HSD enumeration)

11 effective Friday theorems (T2450-T2462 with T2452 atomic-claim burned per calibration note).

Casey-decision items in queue

(Filed in queue_casey.md Friday morning 08:48 EDT)

1. Paper #125 v1.0 venue submission timing
2. Paper #126 v0.2 venue selection (CMP / Inventiones)
3. Paper #127 v0.1 venue selection (JFA / CMP)
4. T2449 multi-CI ratification (Thursday inherited)
5. CI_BOARD.md staleness update
6. Vol 1 v1.0 cover-to-cover authorization
7. Sessions 15-16 pre-spec authorization (Keeper)
8. Friday prompt item #11 (“Single approach without acrobatics paper”) scope clarification

PCAP cadence achievement summary

- 11 effective Friday theorems delivered in ~60 min
- ~5 min/theorem average (PCAP cadence)
- 13 toys all 6/6+ or 7/7 or 8/8 PASS
- Honest-scope refinement applied to T2459 via extended verification (T2462)
- Casey FAST CADENCE directive honored

Filing status

v0.1 roadmap filed Friday 2026-05-22 ~08:50 EDT (date-verified actual).

Pending for v0.2: - Keeper Sessions 15-16 pre-spec formalization - Casey decisions on items 1-8 (queue_casey.md) - Cross-CI absorption of Friday morning Lyra-lane

deliverables

— Lyra, Strong-Uniqueness v0.11+ Path Roadmap v0.1, Friday 2026-05-22 ~08:50
EDT (date-verified actual)